15 December 1975

MILITARY SPECIFICATION

FREEZER BASE; WITH WORK TOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a self-contained two door freezer base with a stainless steel table height work top.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-F-1421	-	Fluorocarbon Refrigerants
HH-I-524	-	Insulated Board, Thermal (Polystyrene)

MILITARY

MIL-V-173	-	Varnish, Moisture and Fungus-Resistant (For Treatment of Communication, Electronic, and Associated Electrical Equipment)
MIL-R-12323	-	Refrigerators and Related Equipment, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	-	Enviromental Test Methods

FSC 4110

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM)

A 167 - Corrosion-Resisting Chromium Steel Plate, Sheet, and Strip

A 506 - Regular Quality Hot-Rolled and Cold-Rolled Alloy-Steel Sheets and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Underwriters' Laboratories, Inc., (UL)

UL - 471 - Commercial Refrigerators

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio St., Chicago, IL 60611; 1285 Walt Whitman Road, Melville, L.I., NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Sanitation Foundation (NSF), Publications

No. 1 - Soda Fountain and Luncheonette Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, MI 48106.)

American National Standards Institute (ANSI)

B38.1 - Food Storage Volume and Shelf Area of Automatic Household Refrigerators, Method of Computing

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10016.)

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National Bureau of Standards (NBS), Publications

H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Director, National Bureau of Standards, Washington, DC 20234.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When required (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

3.2 Standard product. The freezers delivered under this specification shall be a current standard product of the supplier, except for changes necessary to meet specification requirements.

3.3 Codes and standards. The freezers shall comply with the applicable requirements of UL 471 for safety and NSF No. 1 for sanitation.

3.3.1 Compliance. Prior to commencing production the supplier shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the freezers he proposes to supply under this specification meet the applicable requirements of UL 471 and NSF No. 1 as follows:

3.3.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL listing mark or label, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the freezers offered have been tested and conform to UL 471.

3.3.1.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished freezer, or

(2) A certification for the freezer issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the freezers are being procured, indicating that the freezers have been tested and conform to the applicable requirements of NSF No. 1.

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3.4 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for freezers, provided the complete item complies with all provisions of this specification.

3.4.1 Plastics. Plastics used in the freezer shall not impart odor or taste to foods or liquids (see 4.6.1.1). Plastics shall not craze, crack, or permanently distort from temperature or under conditions of normal use (see 4.6.1.2.1), during storage in temperatures ranging from 0 to 140 degrees Fahrenheit (F.) (see 4.6.1.2.2) or due to thermal shock (see 4.6.1.2.3). When foamed-in-place polyurethane insulation is used, all plastics in direct contact with the insulation shall be ABS (acrylonitrile, butadiene, and styrene).

3.4.2 Steel.

3.4.2.1 Stainless steel. Stainless steel sheet or strip shall conform to ASTM A 167.

3.4.3 Insulation. All insulation shall be industrial type fiberglass, expanded polystyrene, or expanded polyurethane, shall be homogeneous throughout, and shall have a "K" factor using a 75 degree fahrenheit (F.) mean temperature and expressed in British Thermal Units (Btu) per hour, per square foot, per degree temperature difference; per inch thickness. Loose fill insulation shall not be acceptable. Flame retardant grades shall be used where needed to meet requirements of UL Standard No. 471.

3.4.3.1 Pre-expanded insulation. Pre-expanded insulation shall be polystyrene material conforming to class A or B, types I or II of HH-I-524.

3.4.3.2 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial rated "K" factor of not more than 0.15, a core density of not less than 1.7 pounds per cubic foot when tested as specified in (4.6.9).

3.4.4 Fasteners. Screws, bolts, nuts, and other fasteners shall be stainless steel or cadmium, or nickel plated low carbon steel. Threads shall conform to Handbook H-28, National Bureau of Standards.

3.4.5 Refrigerant. The refrigerant shall conform to BB-F-1421.

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3.5 Design. The freezer shall be of welded frame construction designed as a self-contained two door, freezer base with a stainless steel work top (see 3.4.2.1), and when specified (see 6.2), a 6-inch high backsplash shall be provided. The freezer shall be designed to stand on four 6-inch adjustable legs or when specified (see 6.2) shall be provided with casters. The freezer shall be equipped with 2-wire shelves, one per opening. Unless otherwise specified (see 6.2), the freezer shall be designed to operate on 115 volt, 1-phase, 60 Hertz electrical system. The refrigeration system shall be air cooled designed to operate on refrigerant specified in 3.4.5. The condensing unit shall be powered by a 1/3 horsepower (hp) compressor, hermetic type thermally or mechanically sealed. The condensing unit shall be capable of being removed for cleaning and repairing. Unless otherwise specified (see 6.2), freezers shall be equipped with automatic defrost. A condensate evaporator shall be furnished for removal of condensate without the use of plumbing or floor drains.

3.5.1 Physical characteristics. The overall maximum dimensions of the freezer base shall be 74 inches long by 36 inches deep by 36 inches high (including legs). When a backsplash is required, the height shall be 42 inches. The minimum capacity of the freezer shall be 14 cubic feet.

3.5.2 Performance. When operating in a 100 F. ambient temperature, the refrigeration system shall be capable of maintaining an average food temperature of 0 F., or lower, without any food temperature exceeding 10 F. when loaded to 70 percent of the required storage volume of the cabinet and with the compressor running time not to exceed 90 percent. The refrigeration system shall also be capable of lowering the empty cabinet temperature from 110 to 0 F. in not more than 8 hours (see 4.6.5). When operating in a 90 F. ambient temperature, the refrigeration system shall be capable of maintaining a temperature of 0 F., or lower, within the cabinet with no load. The compressor running time shall not exceed 80 percent for no load and storage load operation in the 90 F. ambient temperature (see 4.6.5). The freezer shall be capable of safe storage and transportation without permanent impairment in air temperatures as high as 140 F. with no solar radiation and in air temperatures as low as 0 F. (see 4.6.1.2.2).

3.5.3 Automatic defrosting. Freezers with automatic defrosting shall be manually initiated or automatically initiated with automatic reset and shall not require the removal of food during the defrosting period. The defrosting operation shall not raise the average temperature of the frozen food more than 5 F. when the freezer compartment is loaded with food to 70 percent of its required storage capacity. Temperature increases shall be based on the highest average temperature observed during the defrosting period and the average temperature observed just prior to initiation of defrost (see 4.6.3).

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3.5.4 Cabinet. The design and fabrication of the refrigeration system and the cabinet shall prevent water vapor or water passing into the insulated space from the outside or water entering the insulated space from the inside during operation, storage, cleaning, or defrosting.

3.5.5 Doors. The exterior door panels shall be stainless steel of a thickness not less than 18 gage (0.050 inch nominal) in accordance with 3.4.2.1. The gasket on the perimeter shall provide a seal around the doors and the method of attachment shall provide easy gasket replacement. The doors shall be one left hand and one right hand opening in the center. Anti-sweat heaters shall be installed on the perimeter of each door opening for condensation control.

3.5.5.1 Door hardware. Door handles, latches, and hinges shall be surface mounted and designed so as not to protrude beyond the front panel of the door when in the closed position. All screws into metal shall have machine screw threads. Self-tapping sheet metal screws shall not be used. Each door latch, strike assembly and hinge shall be of commercial heavy duty quality and constructed of parts replaceable from suppliers current stock.

3.5.6 Cabinet inner liner. The inner liner of the freezer shall be fabricated of stainless steel specified in 3.4.2.1. An automatic condensate disposal system shall be furnished and the interior shall be completely wired with a light in each compartment controlled by a switch activated by the door.

3.5.6.1 Breaker strips. Breaker strips of thermoplastic material, thermosetting plastic, natural rubber, or synthetic rubber shall be installed to join the outer shell and inner liner around the exterior door openings. Joints between the cabinet inner liner and the bottom breaker strip and the lower 4-inches of the side breaker strips shall be sealed to prevent liquid spillage or defrost water from entering the insulation space. Breaker strips shall not be damaged or displaced by thermal shock (see 4.6.1.2.3). Joints in the breaker strips shall be smooth and cleanable; edges of strips shall be eased or rounded; and the entire installation shall provide a smoothly finished appearance.

3.5.7 Cabinet exterior. The exterior shall be fabricated of stainless steel specified in 3.4.2.1 with a thickness of not less than 18 gage (0.050 inch nominal). The exterior panels and doors shall have a No. 4 finish.

3.5.8 Top. The freezer top shall be flat and constructed of material specified in 3.4.2.1 with a thickness of not less than 16 gage (0.0625 nominal thickness). The top surface shall not flex under a concentrated load (see 4.6.4.1). Joints that are not welded airtight shall be sealed to prevent moisture infiltrating the insulation. The front edge shall be turned down approximately 3-1/8 inches. When specified (see 6.2), the backsplash shall have a coved radius approximately 3/8 inches at turn up and returned approximately 2 inches to conceal and house electrical outlet boxes.

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3.5.9 Insulation installation. The freezer and doors shall be insulated with material specified in 3.4.3. The insulation thickness shall be sufficient to prevent all the exterior surfaces from being more than 10 F. below the room ambient temperature when the freezer is operating (see 4.6.4.2).

3.5.10 Charging of refrigeration system. The refrigeration system shall be charged with refrigerant and oil by the manufacturer. After charging, the refrigeration system shall not freeze up due to moisture in the refrigerant when the freezer is operated (see 4.6.6.1).

3.5.11 Fungus and moisture resistance of electrical components. The freezer shall be treated for fungus and moisture resistance (see 4.6.8). The electric circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

a. Component or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Component or circuit elements whose functions will be adversely affected by the varnish shall not be treated.

3.5.12 Protection against electrolytic corrosion. Freezers shall be constructed to provide protection against electrolytic or galvanic corrosion. Joints between dissimilar metals, which may become wetted in service, shall be protected against electrolytic or galvanic corrosion by proper selection of materials, plating, isolation, insulation, paint, area relationship, gaskets, or other means.

3.5.13 Operation. The freezer shall cycle properly; operate according to control settings; machanical parts shall be in working order; and there shall be no refrigerant leaks when tested as specified in 4.6.1.2.3.

3.6 Marking for identification. The manufacturer's name that normally appears on the outside of his product and any other standard nameplate data shall be placed on all freezers according to the manufacturer's standard practice.

3.7 Wiring diagram. A wiring diagram shall be glued or otherwise affixed to the freezer. The diagram shall show each electrical component by sketch, outline, or symbol, and shall include the name of each component. The wiring diagram shall show the wiring to all components and any coding used for identification of wires. The wiring diagram shall be located in a visible area in the condensing unit section.

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3.8 Workmanship. Welding and brazing shall be complete, uniform and properly fused, having no burn holes, slag inclusions, scale, or flux deposits, and shall not be cracked, fractured, or undercut. Exposed welds shall have discoloring removed. Soldering shall be complete, clean, and adherent, and shall not have pinholes. Bolts, nuts, screws, studs, and other types of threaded fasteners, when used, shall not be broken, stripped, fractured, or loose, and shall have lock washers or shall be self-locking type. Sheet metal or structural sections shall not be fractured, split, punctured, torn, dented, creased, bowed, sprung or malformed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article. When a first article is required, the refrigerator shall be examined for defects listed in table I, dimensions specified, and tested as specified in 4.4.1 thru 4.6.9. The presence of any defect, any dimension not as specified, or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Plastics. The plastic test shall be performed as specified in 4.6.1.1. The inspection lot shall be the amount of each type of plastic offered for inspection at one time. The sample unit shall be one specimen from the lot. Failure to pass the test shall be cause for rejection of the lot.

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4.4.1.2 Foamed-in-place insulation. The density of foamed-in-place insulation shall be tested as specified in 4.6.9 by testing a foamed sample prior to each day's production. Failure to pass the test shall be cause for rejection of the sample. In addition, the foam insulation calibration of mix (mix proportion), amount of flow (per freezer), and time of flow (per freezer) shall be recorded daily.

4.4.2 Process examination. Examination shall be made of the following processes to establish conformance to specified requirements. Whenever non-conformance is noted, correction shall be made to the affected items and process.

(a) Sealing of openings in cabinet shell against moisture infiltration (see 3.4.4).
(b) Inner liner joints sealed (see 3.5.6.1 and 3.5.8).
(c) Installation of proper insulation (see 3.4.3 and 3.5.9).
(d) Installation of ABS plastic when in contact with foamed-in-place insulation (see 3.4.1).
(e) Priming of cabinet exterior (see 3.5.7).
(f) Fungus and moisture resistance treatment (see 3.5.11).
(g) Protection of joined dissimilar metals against corrosion (see 3.5.12).

4.4.3 End item inspection. The inspection lot shall consist of all freezers offered for inspection at one time. The sample unit shall be one complete freezer.

4.4.3.1 Visual examination. Examination shall be made of the freezer for the defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturer's standard product except for changes necessary to comply with specification requirements	X	
Finish	Not finished, plated, or coated (where required)	X	
	Type or color of finish not as specified		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Design, construction and workmanship general (applicable to all components and assemblies)	Fractured, split, punctured, torn, dented, creased, bowed, sprung or malformed	X	
	Part missing, misplaced, loose or not in proper alignment, or not type specified	X	
	Doors are not left and right handed	X	
	Functioning component that is inoperative or will not operate as intended	X	
	Adjustable assembly that cannot be adjusted, or is not properly adjusted to perform the function intended (condition cannot be readily rectified)	X	
	Not properly adjusted but can be readjusted		X
	Gasket not around complete perimeter of each door	X	
	Gasket not easily replaceable	X	
	Cabinet does not contain condenstate outlet	X	
	Cabinet does not contain a light in each compartment	X	
	Joints in breaker strips are smooth		X
	Edges of breaker strips not eased or rounded		X
	Component not accessible for servicing where required	X	
	Welding and brazing missing, having burn holes, cracked, fractured, or not fused	X	
	Welding and brazing having slag inclusions, is undercut, not uniform, with scale or flux deposits not removed		X
	Exposed welding that is discolored		X
	Soldering missing, not adherent	X	
	Soldering not clean (flux or flux residue not removed); pinholes in solder or incomplete	X	
	Bolts, nuts, screws, studs and other types of threaded fasteners missing, broken, stripped, fractured or loose	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Design, construction and workmanship general (applicable to all components and assemblies) (cont'd)	Bolts, nuts, screws, studs and other types of threaded fasteners having lockwashers missing unless self-locking type		X
Wiring diagram	Missing, incomplete, not legible not affixed in the prescribed manner		X
Marking for identification	Missing, incomplete, not legible		X

4.4.3.2 Dimensional examination. The freezer shall be examined for compliance with the dimensions specified. Any dimension that is not within the specified requirements shall be a defect. The inspection level shall be S-2 with an AQL of 6.5, expressed in terms of defects per hundred units.

4.4.3.3 End product testing. Each freezer shall be tested as specified in 4.6.6.2, when applicable, and 4.6.7. Any nonconformance shall be cause for rejection of the unit. When a combination liquid and moisture indicator is furnished with each unit, one production unit from the initial lot shall be tested as specified in 4.6.6.1. Any nonconformance shall be cause for rejection of the lot. When a combination liquid and moisture indicator is not furnished with each unit, the test specified in 4.6.6.1 shall be performed using an inspection level of S-1 and an AQL of 6.5, expressed in terms of defects per hundred units. The tests specified in 4.6.1.2 through 4.6.5, and 4.6.8 and 4.6.9 shall be performed on one production unit when a first article is not specified. Failure of any test shall be cause for rejection of the inspection lot.

4.4.4 Examination of preparation for delivery. Inspection of preparation for delivery shall be performed in accordance with the inspection requirements in MIL-R-12323.

4.5 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements (see 3.3), shall be examined and validated.

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4.6 Tests.

4.6.1 Plastic and cabinet tests.

4.6.1.1 Odor and taste test. Plastic shall be given the following odor and taste test to determine compliance with the odor and taste requirements of 3.4.1. Place the plastic to be tested with approximately 5 cc of distilled water in a 16-ounce, wide-mouth, glass jar. Place a beaker containing 100 to 150 cc of distilled water in a second opened glass jar. The two jars shall be kept at 45 to 50 F. for 24 hours. At the end of this period an examination shall be made for significant odor or taste difference between the two waters. Any odor or taste difference shall constitute failure to comply with 3.4.1 and shall be cause for rejection of the lot. Containers of other shapes, capacities, and types may be substituted for the containers described, if odd shaped pieces of plastic are tested, provided the intent of the odor and taste test is met.

4.6.1.2 Normal use and storage tests.

4.6.1.2.1 Normal use test. The assembled freezer shall be permitted to come to steady state with the interior cabinet temperature at the lowest value obtainable with continuous operation of the refrigerating unit in an ambient temperature of 70 F. to determine compliance with the plastic normal use requirement of 3.4.1. Noncompliance shall constitute failure of this test.

4.6.1.2.2 Storage test. The inoperative freezer without storage load and with the door open shall be permitted to come to thermal equilibrium in ambient temperatures of 0 to 140 F. to determine compliance with the storage requirements of 3.4.1 and 3.5.2. Noncompliance shall constitute failure of this test.

4.6.1.2.3 Thermal shock transportation tests. The freezer shall be placed in an ambient temperature of 140 F. for a period of 12 hours and, during the last hour of the 12 hour period, shall be subjected to the following two tests:

(a) One side of the freezer shall be raised 3 inches off a level concrete floor by a single support which can be withdrawn to allow a free drop. This procedure shall be repeated allowing each of the four sides to drop.

(b) Each door shall be opened 90 degrees and slammed closed 10 times. At the end of the 12 hour period, the freezer shall be transferred, as quickly as practical, to a room maintained at 0 F. The freezer shall remain in this room 12 hours. During the last hour of the 12 hour period, the cabinet shall be resubjected to the two tests (drop and door slamming) described above. During and after these tests, the freezer shall be examined for compliance with the thermal shock requirements of 3.4.1. 3.5.2, and 3.5.6.1. After the freezer

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has returned to room temperature, it shall be operated for 2 hours and examined for compliance with the normal use and operating requirements of 3.4.1 and 3.5.13. The test of 4.6.1.2.2 may be run concurrently with the test in this paragraph. Noncompliance with any requirement shall constitute failure of this test.

4.6.2 Storage capacity test. Determine storage capacity in accordance with ANSI B38.1. Failure of volume to equal a minimum 14 cubic feet (3.5.1) shall constitute failure of the test.

4.6.3 Defrost tests. The freezer with automatic defrosting shall be tested as follows to determine compliance with the defrosting requirements of 3.5.3 and 3.5.4. Noncompliance shall constitute failure of the test.

4.6.3.1 Defrost test conditions. The freezer cabinet shall be loaded with sealed test packages, as described in ANSI B38.1, to 70 percent of its established storage capacity with material having a density of 35 pounds per cubic foot. Commercially packaged chopped spinach may be used in lieu of the material described in ANSI B38.1 provided the spinach has a density of 35 pounds per cubic foot. Ambient conditions for the test shall be 90 F. and a relative humidity between 80 and 85 percent. The temperature control unit shall be adjusted to obtain an average cabinet air temperature of 0 $\pm$ 2 F. Electrical characteristics for operation of this test shall be the same as equipment rating.

4.6.3.2 Procedure. During the first hour of the initial 8 hours of the test, both doors shall be opened wide seven times per hour for 10 second periods at 8 minute intervals, and the procedure shall be repeated during the fourth and and seventh hours. Following the door opening schedule, the freezer shall stabilize for the next 16 hours without doors being opened. The defrosting shall begin immediately following the stabilization period. Automatically initiated defrost systems may be prepared prior to the test to prevent defrosting during the door opening and stabilization phase of the test. The defrost period shall be terminated when the temperature of the freezer cabinet has peaked and the freezer shall then continue to operate over a 4 hour recovery period. The temperature of the freezer cabinet and the test packages shall be taken continuously, or at intervals of not greater than 15 minutes, starting prior to initation of the defrost cycles and continuing throughout the recovery period.

4.6.4 Freezer cabinet tests.

4.6.4.1 Freezer top test. A concentrated load of 50 pounds shall be on 1 square inch area to nine individual locations separated from each by at least twelve inches (see 3.5.8). Any noticeable deflection of the top surface shall constitute failure of the tests.

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4.6.4.2 Exterior surface temperature test. The exterior surface temperature of the freezer cabinet shall be tested to determine compliance with the insulation requirements of 3.5.9. During the performance of the test specified in 4.6.5, when the freezer is operating and the inside temperature is at equilibrium, the entire exterior of the cabinet, including doors and top, shall be inspected to determine that all surfaces are not more than 10 F. below the test room ambient temperature. Inability to conform to the requirements in 3.5.9 shall constitute failure of the test.

4.6.5 Performance tests. The freezer shall be subjected to an average-food temperature test, a storage-load test, a no-load test, and a no-load pulldown test to determine compliance with temperature and compressor running time requirements of 3.5.2. Automatic defrost systems may be shunted out for these tests. The tests shall follow the procedures in ANSI B38.1 with modified conditions as requird by this specification and with the requirement that the average-food temperature test shall be made in accordance with the same procedure outlined for the modified storage load test. The loads for the average-food temperature test and the storage-load test shall be 70 percent of the storage capacity as determined by 4.6.2. The test for no-load pulldown shall be made in accordance with ANSI B38.1 with the thermal equilibrium temperature at completion of test not to exceed 2 F. Average-food temperature test shall be conducted in an ambient temperature of 100 F. The no-load and the storage-load tests shall be made in an ambient temperature of 90 F. The temperature sensing devices in the interior of the freezer cabinet shall be located in each compartment to provide data on average temperature and high and low temperature of each compartment, subject to location requirements specified hereinafter. A compartment is defined as a space, designed for storage of food, bounded by the inner liner, the top of the immediate shelf, and a plane passing vertically through the front edge of the shelf. The temperature sensing devices shall be located inside of each compartment in accordance with ANSI B38.1. Inability to conform to the requirements of 3.5.2 shall constitute failure of the test.

4.6.6 Refrigerant system moisture test.

4.6.6.1 Freeze-up moisture test. The freezer shall be tested as follows to determine compliance with the moisture requirements of 3.5.10. The freezer shall be connected to a source of power and operated continuously for 8 hours in a temperature of 110 F. ambient, the power shall be shut off and the unit allowed to stabilize for 8 hours in an ambient temperature 70 F. Immediately after this 8-hour period, the freezer shall be operated continuously as above for 8 hours in an ambient temperature 70 F. During this period, examination shall be made for evidence of refrigerant system freeze-up. When the combination liquid and moisture indicator is used, it shall indicate "dry" to conform with the requirements of 3.5.10. Any evidence of system freeze-up shall constitute failure of the test.

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4.6.6.2 Indicator moisture test. When a moisture indicator or a combination liquid and moisture indicator is furnished with each unit, the indicator shall be observed after the operation test (4.6.7). A liquid line temperature of not more than 100 F. shall be maintained when observing the indicators. Failure of the indicators to indicate "dry" to conform with the requirements specified in 3.5.10 shall constitute failure of the test.

4.6.7 Operation test. The freezer shall be operated 30 minutes at a normal control setting to determine compliance with 3.5.13. After completion of this portion of the test and with the cabinet temperature at 70 ± 3 F., all accessible refrigerant joints shall be tested for leaks to determine compliance with 3.5.13. A halide torch of high sensitivity, or other approved method shall be used. All refrigerant joints not accessible after assembly of the frozen food cabinet shall be tested by a means and method equivalent to the above test. Indication of a leak shall constitute failure of the test.

4.6.8 Fungus test. The fungus test shall be in accordance with test method 508 of MIL-STD-810 (see 3.5.11).

4.6.9 Insulation density test. One free foamed sample of insulation specified in 3.4.3.2 shall be tested from each production lot. The sample shall be a cube, 8 inches on a side. A cube, 6 inches on a side shall be cut from the center of the sample and weighed. This weight of the 6-inch cube shall be multiplied by 8 to determine compliance with the density requirement of 3.4.3.2. Noncompliance shall be cause for rejection of the sample.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A and commercial. Each freezer shall be preserved and packaged in accordance with the applicable requirements of MIL-R-12323.

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

5.2.1 Levels A, B and commercial. Each freezer, preserved and packaged as specified in 5.1 shall be packed in accordance with the applicable requirements of MIL-R-12323.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the applicable requirements of MIL-R-12323.

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6. NOTES

6.1 Intended use. The freezers covered by this specification are for use in the serving lines or kitchens of Military dining halls.

6.2 Ordering data. Purchasers should select the preferred options allowed herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1 and 6.3).
(c) When electrical requirements are other than specified (see 3.5).
(d) When automatic defrost is not required (see 3.5).
(e) When casters are required (see 3.5).
(f) When 6 inch backsplash is required (see 3.5).
(g) Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Contract data requirements. Any requirements for equipment manuals for the item covered by this specification should include in DD Form 1423 Contract Data Requirements List and cited in the contract.

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Preparing activity:

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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